Mathematical Development of the MINBEE High-Resolution Scheme

1 The NVD Framework and TVD-CBC Criteria

The MINBEE scheme is a custom high-resolution formulation synthesized in Leonard's Normalized Variable Diagram (NVD) to perfectly bridge the highly compressive Superbee limiter and the highly diffusive Minmod limiter.

For an upwind node C, an upstream node U, and a downstream node D, the normalized nodal value and face value are defined as:

$$\tilde{\phi}_C = \frac{\phi_C - \phi_U}{\phi_D - \phi_U}, \qquad \tilde{\phi}_f = \frac{\phi_f - \phi_U}{\phi_D - \phi_U} \tag{1}$$

To remain physical and second-order accurate, the scheme is constrained by the Total Variation Diminishing (TVD) and Convection Boundedness Criterion (CBC) restrictions. In the monotonic region ($0 < \tilde{\phi}_C < 1$), the face value must reside within Sweby's TVD boundaries:

$$\tilde{\phi}_C \leq \tilde{\phi}_f \leq \min(1, 2\tilde{\phi}_C) \tag{2}$$

In the non-monotonic regions ($\tilde{\phi}_C \leq 0$ or $\tilde{\phi}_C \geq 1$), the scheme must fall back to pure upwind ($\tilde{\phi}_f = \tilde{\phi}_C$) to prevent numerical overshoots.

2 Parent Schemes in the Normalized Variable Diagram

To build MINBEE, we first define the limits of the TVD region using the standard Superbee and Minmod schemes.

Superbee (Upper TVD Bound): Hugging the upper compressive bounds:

$$\tilde{\phi}_{f,\text{Superbee}} = \begin{cases} 2\tilde{\phi}_C & \text{if } 0 \leq \tilde{\phi}_C \leq 1/3 \\ 0.5\tilde{\phi}_C + 0.5 & \text{if } 1/3 < \tilde{\phi}_C \leq 0.5 \\ 1.5\tilde{\phi}_C & \text{if } 0.5 < \tilde{\phi}_C \leq 2/3 \\ 1 & \text{if } 2/3 < \tilde{\phi}_C \leq 1 \end{cases} \tag{3}$$

Minmod (Lower TVD Bound): Hugging the lower diffusive bounds:

$$\tilde{\phi}_{f,\text{Minmod}} = \begin{cases} 1.5\tilde{\phi}_C & \text{if } 0 \leq \tilde{\phi}_C \leq 0.5 \\ 0.5\tilde{\phi}_C + 0.5 & \text{if } 0.5 < \tilde{\phi}_C \leq 1 \end{cases} \tag{4}$$

3 Derivation of the MINBEE Scheme

The MINBEE scheme is derived by taking the exact arithmetic mean of the Superbee and Minmod profiles in the NVD framework:

$$\tilde{\phi}_{f,\text{MINBEE}} = \frac{1}{2}\left(\tilde{\phi}_{f,\text{Superbee}} + \tilde{\phi}_{f,\text{Minmod}}\right) \tag{5}$$

Evaluating this average across the piecewise sub-intervals collapses the formulation into an elegant, continuous curve:

$$\tilde{\phi}_{f,\text{MINBEE}} = \begin{cases} \tilde{\phi}_C & \text{if } \tilde{\phi}_C \notin [0,1] \\ 1.75\tilde{\phi}_C & \text{if } 0 \leq \tilde{\phi}_C \leq 1/3 \\ \tilde{\phi}_C + 0.25 & \text{if } 1/3 < \tilde{\phi}_C \leq 2/3 \\ 0.25\tilde{\phi}_C + 0.75 & \text{if } 2/3 < \tilde{\phi}_C \leq 1 \end{cases} \tag{6}$$

3.1 Equivalent Flux Limiter Formulation $\psi(r)$

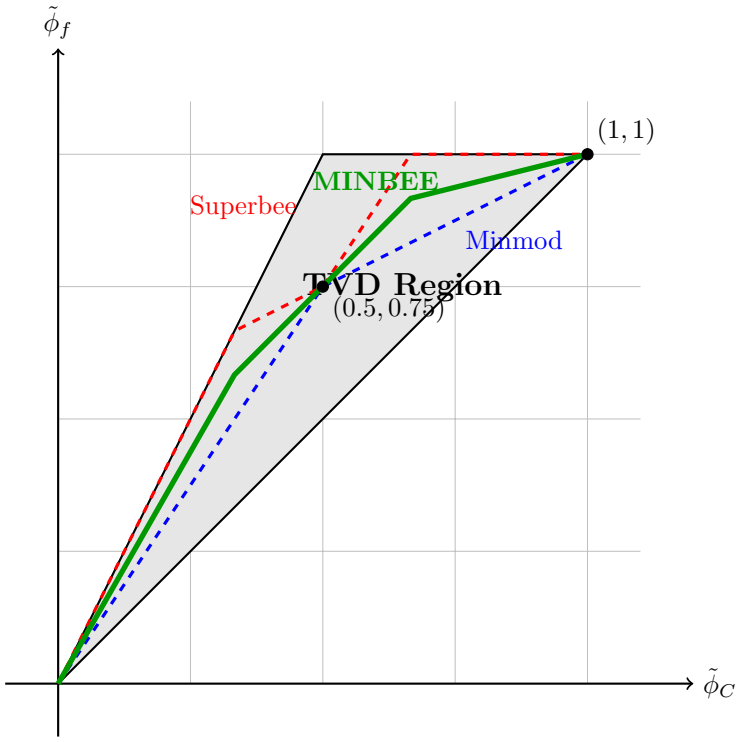
We map the MINBEE NVD formulation back to Sweby's flux limiter framework using the relationship $\psi(r) = 2(\tilde{\phi}_f - \tilde{\phi}_C)/(1 - \tilde{\phi}_C)$ where $r = \tilde{\phi}_C/(1 - \tilde{\phi}_C)$. Evaluating the bounds yields the clean max/min formulation for the MINBEE limiter:

$$\psi(r)_{\text{MINBEE}} = \max\left[0, \min\left(\frac{3}{2}r, \frac{1+r}{2}, \frac{3}{2}\right)\right] \tag{7}$$

4 Graphical Representations

4.1 TVD-NVD Diagram (Normalized Variables)

The following figure illustrates the TVD bounds along with the Superbee, Minmod, and MINBEE schemes inside the Convection Boundedness Criterion (CBC) domain.



4.2 Sweby Diagram (Flux Limiter Region)

The figure below shows the standard Sweby $r - \psi$ plane. The TVD region lies between $\psi(r) = 0$, $\psi(r) = 2r$, and $\psi(r) = 2$.

